

Kaila Raajiv Kiriti Reddy

AI/ML Engineer | Production Pipelines • Deep Learning • Model Deployment

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Professional Summary

AI/ML Engineer with a proven track record of building, deploying, and maintaining production-grade machine learning models and data pipelines. Specialized across the full data lifecycle - from data cleaning and feature engineering to hybrid deep learning architectures, GenAI application development, and MLOps automation. Fluent in Python and major frameworks, with hands-on experience serving models via secure REST APIs and containerized microservices to drive data-driven choices.

Technical Skills

Core ML & Data Science: Python, Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Data Preprocessing, Feature Engineering, Overfitting Mitigation

GenAI & NLP: LLM Application Development, GenAI Prompting, Retrieval-Augmented Generation (RAG), LangChain, LlamaIndex, Hugging Face Transformers

Model Serving & Backend: FastAPI, Microservices, REST APIs, Vector DBs (ChromaDB), SQL, C++

MLOps & Tools: Docker, CI/CD Pipelines, Git (Version Control), Data Drift Detection concepts, Matplotlib, Seaborn

Professional Experience

CR Rao AIMSCS Hyderabad, India

Project Associate - AI Tooling & System Design Nov 2025 - Present

- Served complex ML and GenAI models to production applications by developing scalable, asynchronous FastAPI microservices, improving throughput by 40% while maintaining reproducible Docker deployments.
- Monitored and optimized production workflows, reducing model serving compute costs and runtime latency through multi-GPU deployment pipelines.
- Decreased data processing latency for high-throughput AI workflows by engineering high-performance modern C++ extensions for critical Python processing modules.
- Documented end-to-end model training processes, experiments, and architectural decisions to ensure absolute reproducibility and systemic transparency.

Machine Learning Research Intern Oct 2024 - Oct 2025

- Accelerated model delivery cycles, reducing data preprocessing time by 55%, by constructing scalable Python, Pandas, and SQL data pipelines for clean feature engineering and label validation across large unstructured datasets.
- Achieved 99.18% validation accuracy in sequential pattern recognition by designing, feature engineering, and tuning a hybrid CNN + Bi-LSTM architecture via PyTorch, systematically outperforming baseline classical ML models.
- Resolved severe class imbalances in baseline corpora by programmatically synthesizing 200,000+ validated edge-case training samples using Conditional GANs, ensuring evaluation integrity.

Projects & Applied Research

ResearchMind AI - Production RAG & LLM Platform

- Automated semantic document classification and extraction pipelines across 1,000+ files by architecting an end-to-end platform leveraging autonomous workflows.
- Tuned retrieval performance and context handling by deploying a hybrid RAG pipeline using LangChain, LlamaIndex, and ChromaDB vector stores for precise enterprise search functionality.
- Established robust MLOps observability by containerizing services via Docker and using telemetry tracking to monitor model inference quality, response accuracy, and latency.

Real-Time Model Optimization Engine

- Engineered a lightweight YOLOv8 and PyTorch computer vision inference pipeline optimized for real-time performance on resource-constrained hardware configurations.

- Applied post-training model compression and quantization techniques to minimize storage footprints while strictly maintaining target predictive metrics.

Education

University College of Engineering, Osmania University
Bachelor of Engineering (B.E.) in Artificial Intelligence and Machine Learning
CGPA: 7.8 / 10.0

July 2025
Hyderabad, Telangana